

Adaptive Cruise Controller for RC Car Using STM32

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Abstract—This project documents the design and development of an RC Car is built from the ground up, as an attempt to implement and test the Adaptive Cruise Control System. Two STM32 Nucleo-Boards from STMicroelectronics are used for the development. The STM32L432KC is used for the transmitter and the STM32H723ZG for the RC Car system. A Python User Interface is developed for controlling the RC Car also perform data acquisition in order to analyze and develop the adaptive cruise controller.

The sensor integrated to the RC Car is LIS3DH accelerometer, and Garmin LIDAR Lite V3 to measure the physical data necessary from the RC Car to develop the Adaptive Cruise Controller. The accelerometer was used to determine the velocity of the RC Car, and the LIDAR as an attempt to measure relative distance and velocity of another RC Car ahead. During the test there were challenges to develop the Adaptive Cruise Controller: the velocity calculated by the accelerometer started to drift from the true velocity as time increased due small noise, and the NRF24L01 transmit every 20ms was losing some communication for 1-1.5s which affect velocity calculation and started to drift.

Due to Earth's gravitational influence on the accelerometer, a calibration was attempted but the communication cut still affected to determine if the calibration was effective. Based on the test, the adaptive cruise controller could not be implemented, and implementing more sensors could provide more parameters to accurately calculate the velocity of the RC Car. In the future work, implement more sensor such as the Hall Effect sensor on the wheels and applying the Kalman Filter for sensor can improve velocity accuracy to enable closed-loop speed control.

Index Terms—Adaptive Cruise Control, STM32, PID Controller, LIDAR, NRF24L01, PyQt5, UART.

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B. Transmitter Node

Subsection content.

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IV. PYTHON GRAPHICAL USER INTERFACE

Your GUI section goes here.

V. EXPERIMENTAL RESULTS

Your results go here.

VI. CONCLUSION

Your conclusion goes here.

REFERENCES

- [1] STMicroelectronics, *STM32H723ZG Datasheet*, Rev. 4, 2022.
- [2] Garmin Ltd., *LIDAR-Lite v3 Operation Manual*, 2017.